

Subash Timilsina

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Ph.D. researcher specializing in controllable generative models, unsupervised, semi-supervised, and self-supervised learning, computer vision, and multimodal representation learning. Demonstrated ability to conduct novel research and translate it into production through industry experience. Publications in leading venues including NeurIPS, ICML, IEEE TSP, SPL and ICASSP.

Skills

Languages: Python, C/C++, MATLAB, $\text{\LaTeX}$

ML / AI: Generative models (GANs, VAEs, diffusion), self-supervised, semi-supervised and multi-modal representation learning, causal inference, convex and non-convex optimization

Architectures: Transformers (LLMs, ViT), U-Net, autoregressive models

Frameworks & Tools: PyTorch, NumPy, Scikit-learn, HuggingFace, PyTorch Distributed, Git, Linux, Flask

Foundations: Linear algebra, tensor learning, statistical learning, probability theory, estimation theory, reinforcement learning, convex/non-convex optimization.

Education

Ph.D. in Computer Science (Minor: **Artificial Intelligence**)

Sep 2021 – Jun 2026

Oregon State University, Corvallis, OR

GPA: 3.97/4.00

Coursework: Deep Learning, Machine Learning, Statistical Learning, Reinforcement Learning, Convex Optimization, Linear Algebra for ML and SP, Probability Theory, Estimation Theory.

B.E. in Electronics and Communications Engineering

Oct 2015 – Oct 2019

Pulchowk Engineering Campus, Tribhuvan University, Nepal

(Top 0.8% in nationwide entrance exam)

Professional Experience

Oregon State University, Corvallis, OR

Sep 2021 – Present

Graduate Research Assistant (Advisor: [Xiao Fu](#))

- (Ongoing) Developing a lightweight content-style activation-steering method to reduce toxic and sycophant generations from LLMs while preserving semantic meaning; evaluating on RealToxicityPrompts/ sycophancy datasets.
- Developed differential independence regularization to disentangle content and style across unpaired multi-domain images replacing statistical independence; improved image translation FID ($\approx \downarrow 2.0$ vs. SOTA) and style diversity LPIPS ($\approx \downarrow 0.15$ vs. SOTA) on AFHQ, CelebA-HQ, and Colored MNIST [1].
- Designed a Jacobian-based style encoder regularization on content-style generative model for class-conditioned generation on long-tailed datasets; reduced FID by ($\approx \downarrow 6.0$) vs. GAN-based SOTA on AnimalFacesLT, FlowerLT, CIFAR10/100LT [Project page].
- Developed an identifiable representation learning algorithm to disentangle shared and domain-specific features from unpaired data; improved multilingual word translation (+7%), domain adaptation (+2%), and single-cell analysis (+30%) over baselines [2, 4].
- Developed the maximum-likelihood estimation based loss objective for completing quantized low-rank tensor with dithering learning measurements distribution using GAN / Deep Image Prior methods. Shown application in radio map reconstruction from heavily quantized, minimal measurements [6, 7, 5].

Oregon State University, Corvallis, OR

Sep 2021 – Mar 2026

Graduate Teaching Assistant – EECS Department

- Courses: Programming & Data Structures, Probability & Stochastic Processes, Signals & Systems II, Networks in Computational Biology.
- Developed autograding system on Gradescope; graded assignments, projects, and exams; held office hours. Substitute instructor for Networks in Computational Biology (graduate level course).

Docsumo, Kathmandu, Nepal

Jun 2020 – Jul 2021

Data Scientist

- Developed an OCR pipeline to extract structured data from financial documents (bank statements, invoices, W2, ACORD) using computer vision (line detection, table layout analysis) and NLP (NER, text similarity, semantic matching).

- Designed a hybrid visual–textual encoder improving template-matching accuracy by 5% via few-shot learning; fine-tuned CascadeTabNet with meta-learning for +2% table detection; reduced manual client annotations by 15%.

Pulchowk Campus Robotics Club, Nepal

Aug 2016 – Aug 2019

Software Developer Robotics

- Represented Nepal as a team member at the *Asia-Pacific Robotics Competition* (Japan 2017, Vietnam 2018); worked on robot perception, estimation, planning and kinematics

Pulchowk Campus, Nepal

Oct 2018 – Aug 2019

Undergraduate Research Assistant (Advisor: Sanjeeb Prasad Panday, UGC Collaborative Research Grant)

- Built a 3-wheeled autonomous mobile robot for real-time sound-source-based disaster victim localization using an 8-microphone cubic array and GCC-PHAT time-delay estimation.
- Developed an audio denoiser based on machine learning approach using variational auto-encoder with the spectrogram of the audio.

Select Publications

1. Content-Style Identification via Differential Independence, **Subash Timilsina**, Hoang-Son Nguyen, Sagar Shrestha, Xiao Fu. *ICML*, 2026.
2. Domain Transfer Becomes Identifiable via a Single Alignment, Sagar Shrestha, **Subash Timilsina**, Hoang-Son Nguyen, Xiao Fu. *ICML*, 2026.
3. Distribution Matching via Generalized Consistency Models, Sagar Shrestha, Rajesh Shrestha, Tri Nguyen, **Subash Timilsina**. *Preprint*. [\[PDF\]](#)
4. Identifiable Shared Component Analysis of Unpaired Multimodal Mixtures, **Subash Timilsina**, Sagar Shrestha, Xiao Fu. *NeurIPS*, 2024. [\[PDF\]](#)
5. Domain-Factored Untrained Deep Prior for Spectrum Cartography, **Subash Timilsina**, Sagar Shrestha, Lei Cheng, Xiao Fu. *IEEE Signal Processing Letters*, 2025. [\[PDF\]](#)
6. Quantized Radio Map Estimation Using Tensor and Deep Generative Models, **Subash Timilsina**, Sagar Shrestha, Xiao Fu. *IEEE Transactions on Signal Processing*, 2023. [\[PDF\]](#)
7. Deep Spectrum Cartography Using Quantized Measurements, **Subash Timilsina**, Sagar Shrestha, Xiao Fu. *IEEE ICASSP*, 2023. [\[PDF\]](#)
8. Search Disaster Victims Using Sound Source Localization, Abhish Khanal, Deepak Chand, Prakash Chaudhary, **Subash Timilsina**, Sanjeeb Prasad Panday, Aman Shakya, Rom Kant Pandey. *ISCRAM*, 2020. [\[PDF\]](#)

Reviewing

- Peer reviewed 20+ papers in NeurIPS, ICLR, TSP, OJSP, ICASSP.

Awards & Achievements

- **Nepal Government Scholarship (Top 0.8%)**: Fully funded undergraduate scholarship through nationwide engineering entrance exam.
- **Two awards** at **Asia-Pacific Robotics Competition** – ABU Robocon 2016-2019.
- **First Prize for Best Project** – LOCUS 2018.
- **Best Project Award (2019)**: Top thesis project, Dept. of Electronics and Computer Engineering, Pulchowk Campus.